TRUEHOLD WELLNESS

Retatrutide (LY3437943)

Locked peptide information sheet

Overview

Retatrutide is an investigational once-weekly peptide developed by Eli Lilly for obesity, type 2 diabetes, and related metabolic diseases. It is currently in Phase 3 clinical trials and is not FDA approved.

Retatrutide is often called a triple agonist because it activates three hormone receptor pathways:

- GLP-1 receptor
- GIP receptor
- Glucagon receptor

This is why Retatrutide is discussed as a next-generation metabolic peptide: it combines appetite, glucose-regulation, and energy-expenditure signaling in one molecule.

Basic Information

Item	Value
Generic name	Retatrutide
Development code	LY3437943
Developer	Eli Lilly
Class	Triple hormone receptor agonist
Receptor targets	GLP-1, GIP, glucagon
Administration in trials	Once-weekly subcutaneous injection
Approval status	Investigational; not FDA approved
Research areas	Obesity, type 2 diabetes, metabolic disease, related complications

Mechanism Of Action

Retatrutide combines three metabolic pathways.

GLP-1 Activity

GLP-1 receptor activation is associated with:

- Appetite suppression
- Slower stomach emptying
- Increased satiety

- Reduced food intake

GIP Activity

GIP receptor activation is associated with:

- Improved insulin response
- Glucose-regulation support
- Potential body-composition effects under study

Glucagon Activity

Glucagon receptor activation is the unique third pathway compared with GLP-1-only or GLP-1/GIP approaches.

It is associated with:

- Increased energy expenditure
- Increased fat-oxidation signaling
- Increased metabolic-rate signaling
- Mobilization of stored energy

Researchers believe this third pathway may help explain why Retatrutide has produced very large weight-loss results in clinical trials.

Clinical Weight-Loss Results

Phase 2 Obesity Trial

At 48 weeks, the highest-dose group in published Phase 2 research showed average weight loss of approximately 24.2%.

Phase 3 Obesity Trial: TRIUMPH-1

In Eli Lilly's May 2026 TRIUMPH-1 Phase 3 topline results, adults with obesity or overweight and at least one weight-related comorbidity, without diabetes, showed the following at 80 weeks:

- 4 mg group: average loss of 47.2 lb, or 19.0%
- 9 mg group: average loss of 64.4 lb, or 25.9%
- 12 mg group: average loss of 70.3 lb, or 28.3%

Lilly also reported that 45.3% of participants taking 12 mg achieved at least 30% weight loss.

Diabetes Results

In Phase 3 type 2 diabetes research, Retatrutide showed significant A1C and body-weight reductions. Lilly reported that A1C reductions were approximately 1.7% to 2.0%, depending on dose.

Potential Benefits Being Studied

- Weight loss: strong clinical-trial signal
- Type 2 diabetes: strong clinical-trial signal
- Fatty liver disease and MASH/MASLD: under study
- Sleep apnea: under study
- Cardiovascular and renal outcomes: under study
- Osteoarthritis-related outcomes: under study
- Chronic low back pain: under study

Body Composition Notes

One concern with rapid weight-loss therapies is loss of lean mass. Retatrutide research has shown large reductions in body weight and fat mass, but members should still track nutrition, protein intake, resistance training, strength, and overall function.

Common Side Effects Reported In Trials

Most reported adverse events are gastrointestinal and tend to occur during dose escalation.

Commonly reported:

- Nausea
- Vomiting
- Diarrhea
- Constipation
- Reduced appetite
- Stomach discomfort

Less commonly discussed:

- Dizziness
- Fatigue
- Dysesthesia, or abnormal skin sensations

Comparison To Related Metabolic Drugs

Compound	Main receptor targets
Semaglutide	GLP-1
Tirzepatide	GLP-1 + GIP
Retatrutide	GLP-1 + GIP + glucagon

What Retatrutide Is Not

Retatrutide is not AOD-9604.

AOD-9604 is a fragment of human growth hormone, specifically amino acids 176-191, studied mainly for fat-metabolism and lipolysis discussions. AOD-9604 is not a GLP-1 agonist, not a GIP agonist, and not a glucagon receptor agonist.

Why People Are Excited About Retatrutide

- Strong appetite suppression
- Improved glucose-control signal
- Increased energy-expenditure pathway through glucagon receptor activation
- Increased fat-oxidation signaling
- Greater average weight-loss results than earlier incretin therapies in reported trials

Current Legal Status

As of June 2026, Retatrutide is not FDA approved. Lilly describes Retatrutide as an investigational molecule legally available only to participants in Lilly clinical trials.

Any commercial source claiming to sell pharmaceutical Retatrutide should be viewed with caution because no approved retail product currently exists.

Practical Summary

If Semaglutide is often viewed as a GLP-1 pathway approach and Tirzepatide as a GLP-1/GIP dual pathway approach, Retatrutide is widely discussed as a leading triple-pathway metabolic therapy.

The biggest difference is the addition of glucagon receptor activation, which may increase energy expenditure while GLP-1 and GIP reduce appetite and improve insulin-response biology. Current Phase 3 obesity results suggest average body-weight reductions approaching 28% in some studied populations, making it one of the strongest metabolic therapies reported to date.

What To Track

- Weight
- Waist
- Hunger and cravings
- Energy
- Digestion and comfort
- Exercise consistency
- Protein intake
- Strength and lean-mass preservation habits

Protocol Note

Protocol details are reviewed by the TrueHold team case by case. The bot does not provide dosing, reconstitution, or administration instructions in chat.

Sources

- Eli Lilly / PRNewswire TRIUMPH-1 Phase 3 obesity topline results: https://www.prnewswire.com/news-releases/lillys-triple-agonist-retatrutide-delivered-powerful-weight-loss-in-pivotal-phase-3-obesity-trial-302778859.html
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- Drugs.com Retatrutide status overview: https://www.drugs.com/history/retatrutide.html